# Module 5: Pharmacogene Variant Interpretation: CYP2C9

### 1. Thresholds and calibration

I chose a maximum credible allele frequency of 0.15 because CYP2C9 variants can be clinically important even when they are relatively common. For example, CYP2C9*2 (rs1799853, AF 0.088) and *3 (rs1057910, AF 0.049) are reduced-function alleles that affect warfarin dosing. Using a typical rare-disease threshold would exclude these important variants. With my 0.15 threshold, only 2 of 41 variants were removed, and none were classified as P/LP or drug-response variants. In contrast, a 0.001 threshold would have removed 10 variants, including four drug-response alleles: R144C (*2), I359L (*3), R150H (AF 0.0158), and R335W (AF 0.0070). This shows why a higher frequency threshold is more appropriate for CYP2C9, since common variants can still have important effects on drug metabolism.

I chose CADD for calibration because it had the highest coverage (92.7%) and could score different variant types, including synonymous and intronic variants. REVEL had lower coverage (80.5%) because it mainly scores missense variants. However, the calibration results were not reliable because only 8 variants had clear ClinVar labels (1 P/LP and 7 B/LB). All five CADD bands contained fewer than 10 variants (Table 1). For example, the (0, 10] band had an LR+ of 2.33, suggesting supporting pathogenic evidence, but this was based on only one P/LP variant. Since the sample size was so small, I could not confidently use these likelihood ratios to classify variants. Therefore, I kept all 22 VUS unchanged rather than making classifications based on unreliable evidence.

| CADD band | n | P/LP | B/LB | LR+ |
|---|---|---|---|---|
| (0, 10] | 4 | 1 | 3 | 2.33 |
| (10, 20] | 1 | 0 | 1 | 0.00 |
| (20, 25] | 3 | 0 | 3 | 0.00 |
| (25, 30] | 0 | 0 | 0 | undefined |
| (30, 50] | 0 | 0 | 0 | undefined |

### 2. Coverage gaps and the regulatory genome

REVEL did not provide scores for 8 of 41 records, including 3 synonymous, 2 intronic, 2 unannotated, and 1 frameshift record. One example is chr10:g.94949282del (ClinVar 285601, B/LB, gnomAD AF 0.0031), which had no score from CADD, REVEL, or AlphaMissense. The Atlas also could not evaluate it because the query only accepts SNVs. However, missing scores

do not mean that the variant is benign. Its B/LB classification comes from ClinVar, not from the prediction tools.
Since consequence_all also includes splice_region_variant, I would use a minigene assay to test whether this deletion affects splicing. I would compare the reference and deletion sequences and analyze the resulting transcripts using RT-PCR. If the benign classification is correct because normal splicing is preserved, I would expect the deletion construct to produce mostly normally spliced transcripts rather than abnormal splice products. This would help determine whether the variant affects RNA processing, although enzyme activity would still need to be tested separately.
The two unannotated records, chr10:g.94942290C>C and chr10:g.94981296A>A, have identical reference and alternate alleles. They appear to represent allele-level entries rather than individual SNVs, although their exact star-allele identities have not been confirmed. Therefore, these records should not be treated as ordinary variants that the prediction tools failed to score.

The Atlas scored 36 variants. Most scores were driven by ALPHAMISSENSE (31 variants), followed by CACTUS_241_WAY (3 variants) and MERGED_SPLICING (2 variants). One interesting example was rs370407644 (E274Q), which had an AVI score of 23.41 driven by splicing, while its REVEL score was only 0.144. This suggests that the Atlas may detect possible splicing effects that REVEL does not capture. However, no variants met the atlas_only rule, so the Atlas mostly provided information similar to the existing missense predictors. For E274Q, I would use RT-PCR or RNA sequencing to check whether the predicted splicing effect actually changes the transcript.
For the Atlas analysis, I also corrected the merge step from rsid to hgvs_g because missing rsids were causing duplicated rows.

**3. A discordant variant, read mechanistically**

I selected rs28371686 (D360E), which was flagged by the revel_vs_am rule. This variant had a low REVEL score (0.082) but a relatively high AlphaMissense score (0.7481). Its CADD score was 23.5, AVI was 21.42 (driven by ALPHAMISSENSE), gnomAD AF was 0.0032, and ClinVar classified it as B/LB. I think the most likely explanation is that D360E affects CYP2C9 enzyme activity without necessarily causing a disease. AlphaMissense suggests that the substitution may affect an important region of the protein, while the low REVEL score suggests a lower likelihood of a broadly damaging missense effect. However, neither score directly measures how well CYP2C9 metabolizes drugs. Similarly, the ClinVar B/LB label does not necessarily

mean that the variant has no effect on drug response, since disease pathogenicity and pharmacogenetic function are different questions.

This explanation fits better than assuming the ClinVar classification is incorrect. A variant can be benign in terms of disease risk but still affect drug metabolism. I also would not completely dismiss the AlphaMissense result because its high score, together with CADD 23.5, suggests that the substitution may have a functional effect. The relatively low allele frequency (0.0032) does not rule this out, since pharmacogenetic variants do not need to be common to affect drug response.

Importantly, D360E corresponds to CYP2C9*5, a known decreased-function allele reported in the literature (AMP CYP2C9 allele recommendations, 2019; Dickmann et al., 2001). This supports the interpretation that D360E can reduce CYP2C9 activity even though the prediction tools disagree. Based on this, I would expect D360E to show reduced S-warfarin metabolism compared with wild-type CYP2C9. However, this functional interpretation comes from previous studies rather than my computational results alone.

**4. What I could not determine**

Based on my analysis, I would keep D360E as unresolved rather than reclassifying it using prediction scores alone. One limitation is that the dataset does not provide information about zygosity or phase. For example, a patient carrying D360E together with another reduced-function CYP2C9 allele may have a different drug response than someone carrying D360E alone. Without patient-level genotype information, I cannot determine its overall effect on drug metabolism.

Another limitation is that the prediction scores do not directly measure enzyme activity. Even though AlphaMissense suggests a possible functional effect, it cannot tell us how much CYP2C9 activity is reduced or whether the effect is the same for different drugs. Other factors, such as co-medications, could also influence the actual drug response.

To investigate this, I would compare D360E and wild-type CYP2C9 using a recombinant enzyme assay measuring S-warfarin 7-hydroxylation. I would calculate intrinsic clearance (CLint = Vmax/Km) to compare their catalytic activity. If D360E consistently showed 50% or less of the wild-type intrinsic clearance, I would consider this strong evidence of reduced enzyme function. On the other hand, if its activity was similar to wild type, I would reconsider my interpretation of the AlphaMissense result. This would help determine the functional effect of D360E, although additional clinical data would still be needed to evaluate its implications for warfarin dosing.